

Homework (Jalapeno)

- Q1.** The ancient Egyptians used the equation $y = x^2 + 2x + 1$ to model the trajectory of projectiles. What is the vertex of this parabola?
- (A) (0, 1)
(B) (-1, 0)
(C) (-1, -1)
(D) (1, 1)
(E) (1, 3)
- Q2.** The axis of symmetry for the parabola $y = x^2 - 4x + 3$ is $x =$
- (A) -1
(B) 0
(C) 1
(D) 2
(E) 3
- Q3.** The ancient Chinese used the parabola $y = -x^2 + 4x + 1$ to model population growth. What is the maximum value of this parabola?
- (A) 5
(B) 4
(C) 3
(D) 2
(E) 1
- Q4.** The graph of $y = x^2 - 2x - 3$ has x -intercepts at $x =$
- (A) -1, 3
(B) 1, -3
(C) -1, -3
(D) 1, 1
(E) 3, 3
- Q5.** The ancient Greeks used the equation $y = x^2 + 2$ to model the shape of a shield. What is the y -intercept of this parabola?
- (A) (0, 0)
(B) (0, 1)
(C) (0, 2)
(D) (0, 3)
(E) (0, 4)
- Q6.** The parabola $y = -x^2 + 1$ opens
- (A) upward
(B) downward
(C) leftward
(D) rightward
(E) not at all
- Q7.** The ancient Romans used the equation $y = x^2 - 4$ to model the shape of a dome. What is the minimum value of this parabola?
- (A) -4
(B) -3
(C) -2
(D) -1
(E) 0
- Q8.** The axis of symmetry for the parabola $y = x^2 + 2x - 3$ is $x =$
- (A) -1
(B) 0
(C) 1
(D) -3
(E) 3

Open Ended Questions (FRQ)

- Q9.** Graph the parabola $y = x^2 + 2x - 1$ and identify its vertex, axis of symmetry, and x -intercepts. Show your work and explain your reasoning.
- Q10.** The ancient Egyptians used the parabola $y = -x^2 + 4x + 3$ to model the trajectory of projectiles. Find the maximum value of this parabola and explain its significance in the context of projectile motion. Show your work and explain your reasoning.

Chef's Tips

- Tip 1: Emphasize the importance of identifying the vertex and axis of symmetry when graphing parabolas.
- Tip 2: Encourage students to use historical and ancient contexts to make the material more engaging and relevant.
- Tip 3: Remind students to show their work and explain their reasoning when solving free-response questions.